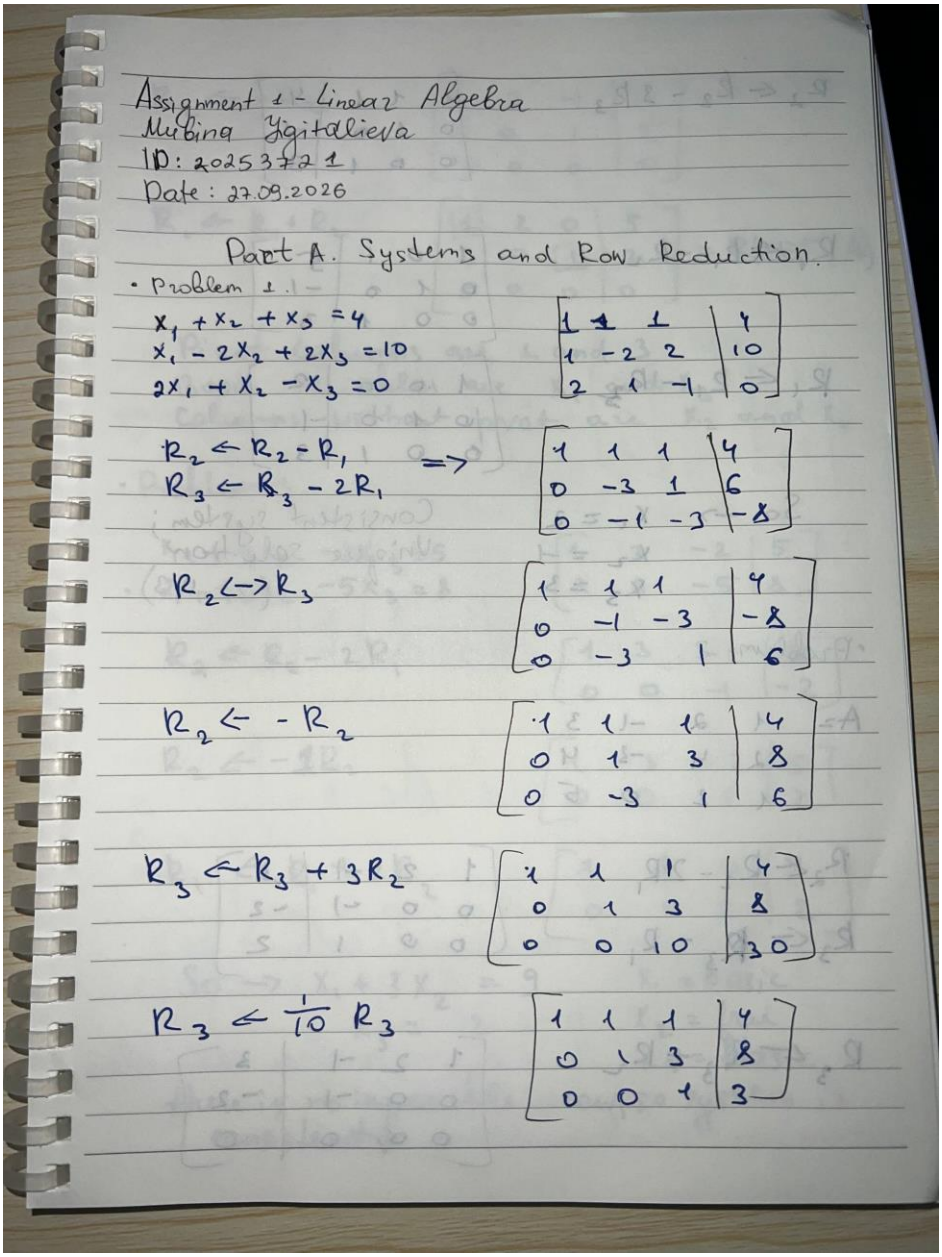
Name: Yigitalieva Mubina

Student ID: 20253721



Assignment 1 - Linear Algebra
Mubina Yigitalieva
ID: 20253721
Date: 27.09.2026

Part A. Systems and Row Reduction.

• Problem 1.

$x_1 + x_2 + x_3 = 4$
$x_1 - 2x_2 + 2x_3 = 10$
$2x_1 + x_2 - x_3 = 0$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 1 & -2 & 2 & 10 \\ 2 & 1 & -1 & 0 \end{array}\right]$$

$R_2 \leftarrow R_2 - R_1$
$R_3 \leftarrow R_3 - 2R_1$ $\Rightarrow$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & -3 & 1 & 6 \\ 0 & -1 & -3 & -8 \end{array}\right]$$

$R_2 \leftrightarrow R_3$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & -1 & -3 & -8 \\ 0 & -3 & 1 & 6 \end{array}\right]$$

$R_2 \leftarrow -R_2$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & 8 \\ 0 & -3 & 1 & 6 \end{array}\right]$$

$R_3 \leftarrow R_3 + 3R_2$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & 8 \\ 0 & 0 & 10 & 30 \end{array}\right]$$

$R_3 \leftarrow \frac{1}{10} R_3$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & 8 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

$R_2 \leftarrow R_2 - 3R_3$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

$R_1 \leftarrow R_1 - R_2$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 5 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

$R_1 \leftarrow R_1 - R_3$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

So → $x_1 = 2$
$x_2 = -1$
$x_3 = 3$

Consistent system; Unique solution $(2, -1, 3)$.

• Problem 2.

$$A = \begin{array}{cccc} 1 & 2 & -1 & 3 \\ 2 & 4 & -3 & 4 \\ 1 & 2 & 0 & 5 \end{array}$$

$R_2 \leftarrow R_2 - 2R_1$

$R_3 \leftarrow R_3 - R_1$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$R_3 \leftarrow R_3 + R_2$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$R_2 \leftarrow -R_2$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$R_1 \leftarrow R_1 + R_2$

$$\left[\begin{array}{ccc|c} \boxed{1} & 2 & 0 & 5 \\ 0 & 0 & \boxed{1} & 2 \\ 0 & 0 & 0 & 0 \end{array}\right] = RREF(A)$$

Pivot columns are 1 and 3
Basic variables are x_1 and x_3
Columns without pivot are x_2 and x_4

- Problem 3.

$x_1 + 3x_2 - 2x_3 = 5$
$2x_1 + 6x_2 - 5x_3 = 8$

$$\left[\begin{array}{ccc|c} 1 & 3 & -2 & 5 \\ 2 & 6 & -5 & 8 \end{array}\right]$$

$R_2 \leftarrow R_2 - 2R_1$

$$\left[\begin{array}{ccc|c} 1 & 3 & -2 & 5 \\ 0 & 0 & -1 & -2 \end{array}\right]$$

$R_2 \leftarrow -1R_2$

$$\left[\begin{array}{ccc|c} 1 & 3 & -2 & 5 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$R_1 \leftarrow R_1 + 2R_2$

$$\left[\begin{array}{ccc|c} 1 & 3 & 0 & 9 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

So $\rightarrow$ $x_1 + 3x_2 = 9$ $\quad x_1 = \text{basic}$
$x_3 = 2$ $\quad x_2 = \text{free}$
$x_3 = \text{basic}$

there is no impossible row, so system is consistent

$x_2 = t$

$x_1 = 9 - 3t$

$(x_1, x_2, x_3) = (9 - 3t, t, 2)$

Parametric form: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 9-3t \\ t \\ 2 \end{bmatrix}$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 9 \\ 0 \\ 2 \end{bmatrix} + t \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$$

- Consistent
- Infinitely many solutions.

- Problem 4.

$x_1 + 3x_2 = 2$

$2x_1 + h \cdot x_2 = k$

$$\left[\begin{array}{cc|c} 1 & 3 & 2 \\ 2 & h & k \end{array}\right]$$

$R_2 \leftarrow R_2 - 2R_1$

$$\left[\begin{array}{cc|c} 1 & 3 & 2 \\ 0 & h-6 & k-4 \end{array}\right]$$

Case 1: Unique Solution

- need a pivot in the second column, so:

$h - 6 \neq 0$

$h \neq 6$ → no free variables.

k can be any real number

- unique solution when $h \neq 6$

Case 2: $h = 6 \rightarrow$ No solution.

If $h = 6$

$$\left[\begin{array}{cc|c} 1 & 3 & 2 \\ 0 & 0 & k-4 \end{array}\right] \quad k \neq 4$$

$h = 6$, $k \neq 4$ gives no solution. inconsistent

Case 3: Infinitely many solutions.

If $h = 6$ and $k = 4$, then second row: $[0 \ 0 \mid 0]$ which means $0 = 0$ no restriction.

No solution: $h = 6$, $k \neq 4$
Unique Solution: $h \neq 6$, any k
Infinitely many solutions: $h = 6$, $k = 4$

Part B. Vector Equations and $Ax = b$

• Problem 5.

$a_1 = (1, 2, -1)$ $b = (1, 3, -4)$
$a_2 = (2, 3, 1)$

Span: $x_1 a_1 + x_2 a_2 = b$

$$x_1 \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \Big| \begin{bmatrix} 1 \\ 3 \\ -4 \end{bmatrix}$$

$x_1 + 2x_2 = 1$
$2x_1 + 3x_2 = 3$
$-x_1 + x_2 = -4$

$x_1 = 1 - 2x_2$

$2(1 - 2x_2) + 3x_2 = 3$
$2 - 4x_2 + 3x_2 = 3$
$2 - x_2 = 3$
$-1 = x_2$
$x_2 = -1$

$x_1 = 1 - 2 \cdot (-1)$
$x_1 = 3$

$-x_1 + x_2 = -4$
$-3 - 1 = -4$ ✓ system is consistent, b span of a_1 and a_2.

$b = 3a_1 - a_2$
$= 3(1, 2, -1) - (2, 3, 1)$
$= (3, 6, -3) - (2, 3, 1)$
$= (1, 3, -4)$

$b \in \text{span}\{a_1, a_2\}$ and $b = 3a_1 - a_2$

• Problem 6.

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 3 \\ 1 & 1 & 1 \end{bmatrix} \qquad x = (1, 2, -1)$$

a) $Ax = x_1 a_1 + x_2 a_2 + x_3 a_3$

$$Ax = 1\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + 2\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} =$$

$$= \begin{bmatrix}1\\2\\1\end{bmatrix} + \begin{bmatrix}0\\2\\2\end{bmatrix} - \begin{bmatrix}2\\3\\1\end{bmatrix} = \begin{bmatrix}-1\\1\\2\end{bmatrix}$$

$A_x = (-1, 1, 2) \rightarrow A_x = a_1 + 2a_2 - a_3$

b) Do the columns span R^3?

Row-reduce A: $\begin{bmatrix}1 & 0 & 2\\2 & 1 & 3\\1 & 1 & 1\end{bmatrix}$

$R_2 \leftarrow R_2 - 2R_1$
$R_3 \leftarrow R_3 - R_1$
$\begin{bmatrix}1 & 0 & 2\\0 & 1 & -1\\0 & 1 & -1\end{bmatrix}$

$R_3 \leftarrow R_3 - R_2$
$\begin{bmatrix}\boxed{1} & 0 & 2\\0 & \boxed{1} & -1\\0 & 0 & 0\end{bmatrix}$

For 3x3 matrix to span R^3, the matrix must have a pivot in every row.

Answer: The columns of A do not span R^3.

Part C. Solution Sets and Independence

• Problem 7.

$$\begin{bmatrix} 1 & 2 & -1 & 1 \\ 2 & 4 & -1 & 3 \end{bmatrix}$$

a) $Ax = 0$

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 0 \\ 2 & 4 & -1 & 3 & 0 \end{array}\right]$$

$R_2 \leftarrow R_2 - 2R_1$

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{array}\right]$$

$R_1 \leftarrow R_1 + R_2$

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 2 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{array}\right]$$

x_1 = basic
x_2 = free
x_3 = basic
x_4 = free

x_2, x_4 are free

Let $x_2 = s$
$x_4 = t$

first row: $x_1 + 2x_2 + 2x_4 = 0$
second row: $x_3 + x_4 = 0 \rightarrow x_3 = -t$

~~first row:~~ $x_1 = -2x_2 - 2x_4$
$x_1 = -2s - 2t$

$$x = \begin{bmatrix} -2s - 2t \\ s \\ -t \\ t \end{bmatrix}.$$

$$x = s\begin{bmatrix}-2\\1\\0\\0\end{bmatrix} + t\begin{bmatrix}-2\\0\\-1\\1\end{bmatrix} \quad s,t \in \mathbb{R}$$

b) $Ax = B$

$$B = \begin{bmatrix}1\\3\end{bmatrix}$$

$$\left[\begin{array}{cccc|c}1 & 2 & -1 & 1 & 1\\2 & 4 & -1 & 3 & 3\end{array}\right]$$

$R_2 \leftarrow R_2 - 2R_1$

$$\left[\begin{array}{cccc|c}1 & 2 & -1 & 1 & 1\\0 & 0 & 1 & 1 & 1\end{array}\right]$$

$R_1 \leftarrow R_1 + R_2$

$$\left[\begin{array}{cccc|c}1 & 2 & 0 & 2 & 2\\0 & 0 & 1 & 1 & 1\end{array}\right]$$

So, $x_1 + 2x_2 + 2x_4 = 2$

$x_3 + x_4 = 1$

Let $x_2 = s$ $\quad x_1 = 2 - 2s + 2t$

$x_4 = t$ $\quad x_3 = 1 - t$

$$x = \begin{bmatrix}2-2s-2t\\s\\1-t\\t\end{bmatrix}$$

$$x = \begin{bmatrix}2\\0\\1\\0\end{bmatrix} + s\begin{bmatrix}-2\\1\\0\\0\end{bmatrix} + t\begin{bmatrix}-2\\0\\-1\\1\end{bmatrix} \quad s,t \in \mathbb{R}$$

the first vector $p = \begin{bmatrix} 2 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ is particular solution.

$x = p + \text{span}\left\{\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ -1 \\ 1 \end{bmatrix}\right\}$ with $p = (2, 0, 1, 0)$

• Problem 8.

a) $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & 2 \\ 2 & 3 & 3 \end{bmatrix}$ $Ax = 0$

$R_2 \leftarrow R_2 - R_1$
$R_3 \leftarrow R_3 - 2R_1$

$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$

$R_3 \leftarrow R_3 - R_2$

$\begin{bmatrix} \boxed{1} & 2 & 1 \\ 0 & -1 & \boxed{1} \\ 0 & 0 & 0 \end{bmatrix}$

only 2 pivots for 3 variables.
one variable is free.

$\{v_1, v_2, v_3\}$ is linearly independent.

Dependence relation:

$c_1v_1 + c_2v_2 + c_3v_3 = 0$

$c_1 = -3 \quad c_2 = 1 \quad c_3 = 1$

$-3v_1 + v_2 + v_3 = 0$

$-3(1,1,2) + (2,1,3) + (1,2,3) =$
$= (-3,-3,-6) + (2,1,3) + (1,2,3) = (0,0,0)$

Answer: Dependent.

b) $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

$R_3 \leftarrow R_3 - R_1 \quad \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$

$R_3 \leftarrow R_3 - R_2 \quad \begin{bmatrix} \boxed{1} & 0 & 1 \\ 0 & \boxed{1} & 1 \\ 0 & 0 & \boxed{-2} \end{bmatrix}$

- 3 variables
- 3 pivot columns
- no free variables.

$Ax = 0$
$x_1 = x_2 = x_3 = 0$

$\{v_1, v_2, v_3\}$ is linearly independent

Part D. Linear transformations and Applications.

Problem 9.

a) $T(x_1, x_2) = (x_1 + x_2, x_2^2)$

For a linear transformation, we must have:

$$T(cu) = cT(u)$$

Take $u = (0, 1)$ $c = 2$

Then $T(u) = T(0,1) = (1, 1)$

so $2T(u) = (2, 2)$

But $T(2u) = T(0, 2) = (2, 4)$

Therefore, $T(2u) \neq 2T(u)$

So, T is not linear transformation.

b) $T(x_1, x_2) = (x_1 - 2x_2, 3x_1 + x_2)$

Standard Matrix:

$A = [T(e_1)\ T(e_2)]$ where $e_1 = (1, 0)$

$e_2 = (0, 1)$

$T(e_1) = T(1,0) = (1-2(0), 3(1)-0) = (1,3)$

$T(e_2) = T(0,1) = (0-2(1), 3(0)+1) = (-2,1)$

$$A = \begin{bmatrix} 1 & -2 \\ 3 & 1 \end{bmatrix}$$

Finding pivots.

$$\begin{bmatrix} 1 & -2 \\ 3 & 1 \end{bmatrix}$$ Row-reduce →

$R_2 \leftarrow R_2 - 3R_1$ $\begin{bmatrix} \boxed{1} & -2 \\ 0 & \boxed{7} \end{bmatrix}$

pivot in every row and column. So →

T is onto

T is one-to-one.

• Problem 10.

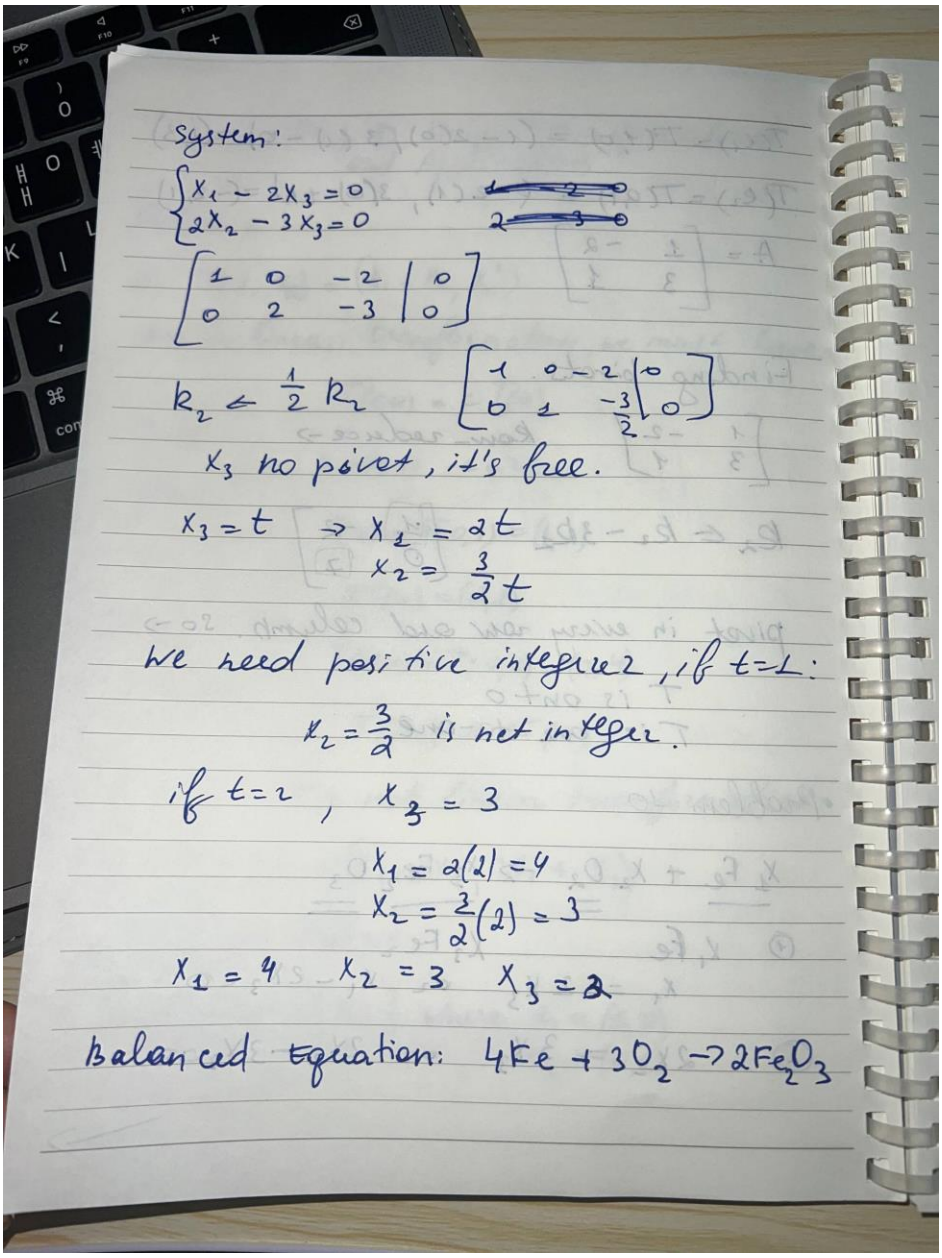
$X_1 Fe + X_2 O_2 \rightarrow X_3 Fe_2 O_3$

① $X_1 Fe$ $X_3 Fe_2$

$X_1 = 2X_3$ or $X_1 - 2X_3 = 0$

② $2X_2 = 3X_3$ or $2X_2 - 3X_3 = 0$

→



System:

$$\begin{cases} x_1 - 2x_3 = 0 \\ 2x_2 - 3x_3 = 0 \end{cases}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 2 & -3 & 0 \end{array}\right]$$

$k_2 \leftarrow \frac{1}{2} k_2$

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & -\frac{3}{2} & 0 \end{array}\right]$$

x_3 no pivot, it's free.

$x_3 = t \rightarrow x_1 = 2t$

$x_2 = \frac{3}{2}t$

We need positive integers, if $t=1$:

$x_2 = \frac{3}{2}$ is not integer.

if $t=2$, $x_3 = 3$

$x_1 = 2(2) = 4$

$x_2 = \frac{3}{2}(2) = 3$

$x_1 = 4 \quad x_2 = 3 \quad x_3 = 2$

Balanced Equation: $4Fe + 3O_2 \rightarrow 2Fe_2O_3$

BONUS

Using conservation of flow:
flow in = flow out

A: 60 enters
B: 40 leaves
C: 20 leaves

$x_1 : A \to B$ $x_2 : B \to C$ $x_3 : C \to A$

At node A:

$$60 + x_3 = x_1$$
$$x_1 - x_3 = 60$$

At node B:

$$x_1 + 40 = x_2$$
$$x_1 - x_2 = 40$$

At node C:

$$x_2 + 20 = x_3$$
$$x_2 - x_3 = 20$$

$$\begin{cases} x_1 - x_3 = 60 \\ x_1 - x_2 = 40 \\ x_2 - x_3 = 20 \end{cases} \qquad \left[\begin{array}{ccc|c} 1 & 0 & -1 & 60 \\ 1 & -1 & 0 & 40 \\ 0 & 1 & -1 & 20 \end{array}\right]$$

$R_2 \leftarrow R_2 - R_1$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 60 \\ 0 & -1 & 1 & -20 \\ 0 & 1 & -1 & 20 \end{array}\right]$$

$R_2 \leftarrow -R_2$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 60 \\ 0 & 1 & -1 & 20 \\ 0 & 1 & -1 & 20 \end{array}\right]$$

$R_3 \leftarrow R_3 - R_2$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 60 \\ 0 & 1 & -1 & 20 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

no pivot in 3rd column

Let $x_3 = t$, then: $x_1 = 60 + t$
$x_2 = 20 + t$

$(x_1, x_2, x_3) = (60 + t, 20 + t, t)$

$t \geq 0$ for nonnegative flows.

system: infinitely many solutions.